INTRODUCTION TO DATA STRUCTURES

MIT SCHOOL OF COMPUTING

AS PART OF CURRICULAM

FIRST YEAR (SEM-2)

COMPUTER SCIENCE AND ENGINEERING

UNDER THE GUIDANCE OF

Prof. Dr. Sachin Tiwari

FOR THE ACADEMIC YEAR 2025-26

SUBMITTED BY

Aditya Satish Lavhande

Subject Teacher	**Course Coordinator**
Prof. Dr. Sachin Tiwari	Prof. Dr. Umesh Nanaware

EXPERIMENT DETAILS

Experiment Number	Experiment Name
1	Arithmetic Operations
2	Built-in Functions
3	Loops

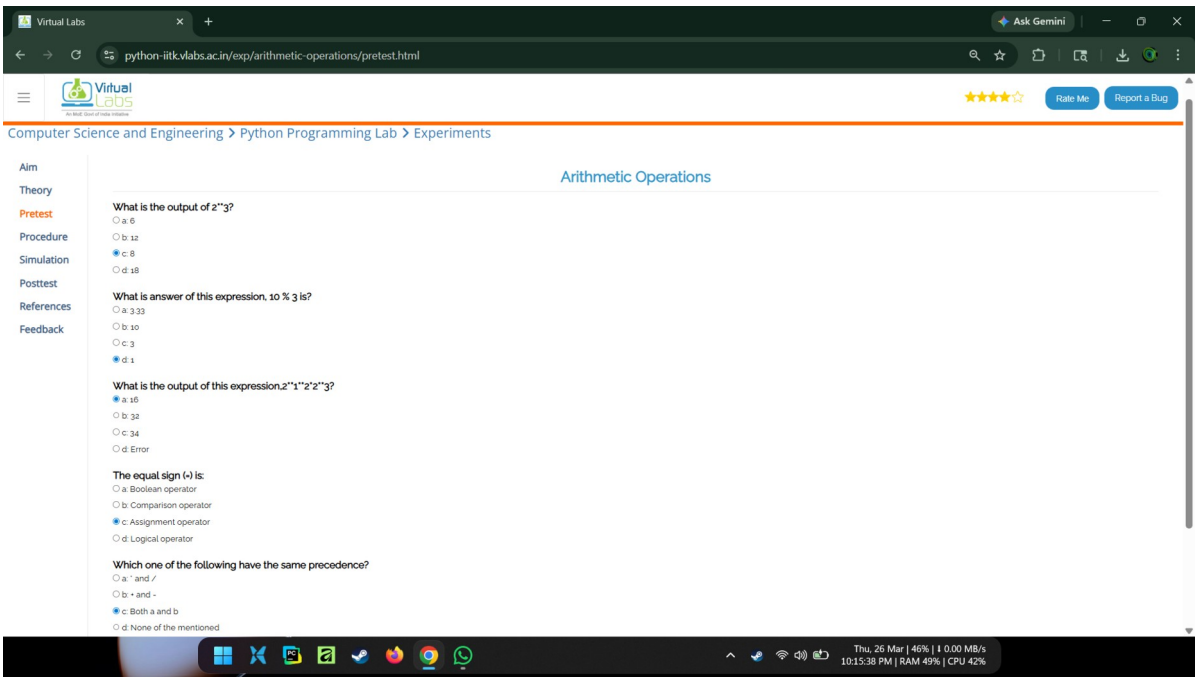
Virtual Lab Name: Virtual Labs

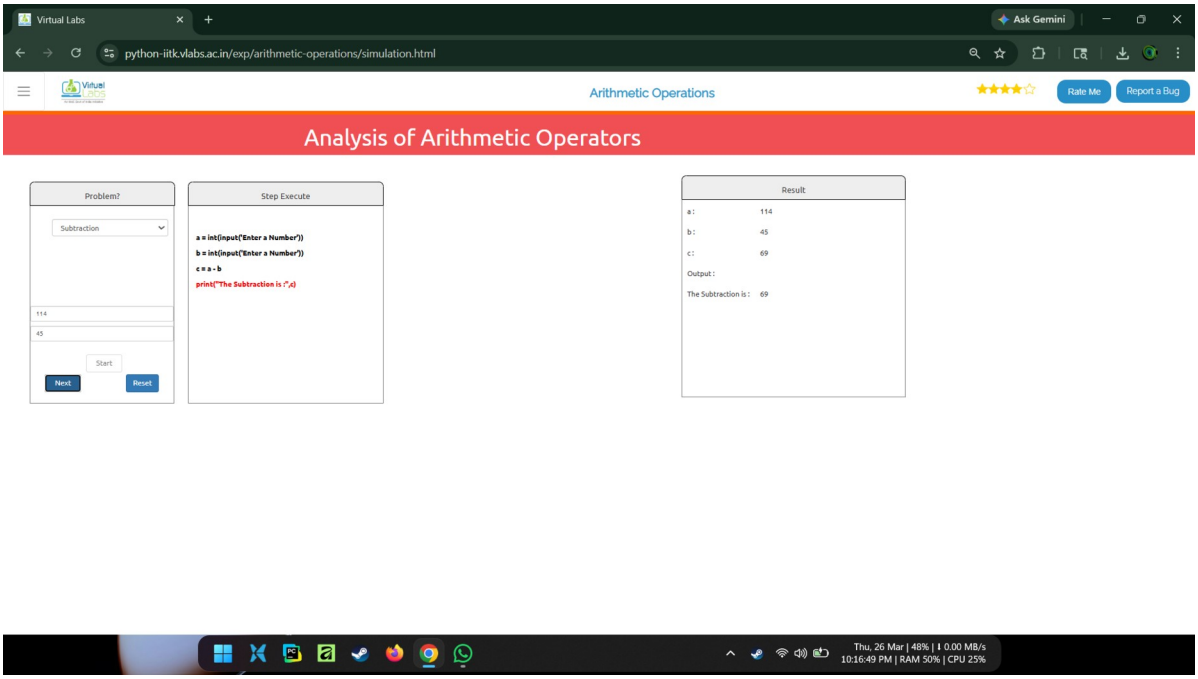
Institute: IIIT-H(Hyderabad)

Date of Performance: 26 March 2026

SCREENSHOTS

Arithmetic Operations





Virtual Labs

python-iitk.vlabs.ac.in/exp/arithmetic-operations/posttest.html

Ask Gemini

Rate Me | Report a Bug

Computer Science and Engineering > Python Programming Lab > Experiments

- Aim
- Theory
- Pretest
- Procedure
- Simulation
- Posttest
- References
- Feedback

Arithmetic Operations

Is Python a case sensitive language?

- a. No
- b. Yes

What will be the output of the following code?

```
x= 2,7
print(x)
```

- a. 2
- b. 7
- c. (2,7)
- d. Error

What will be the output of the following code?

```
x=5.5
y=5.5
print(x+y)
```

- a. 11
- b. 11.0
- c. 10
- d. 0

What will be the output of the following code?

```
a=25.0
b=5.0
print(a%b)
```

- a. 5
- b. 5.0
- c. 0
- d. 0.0

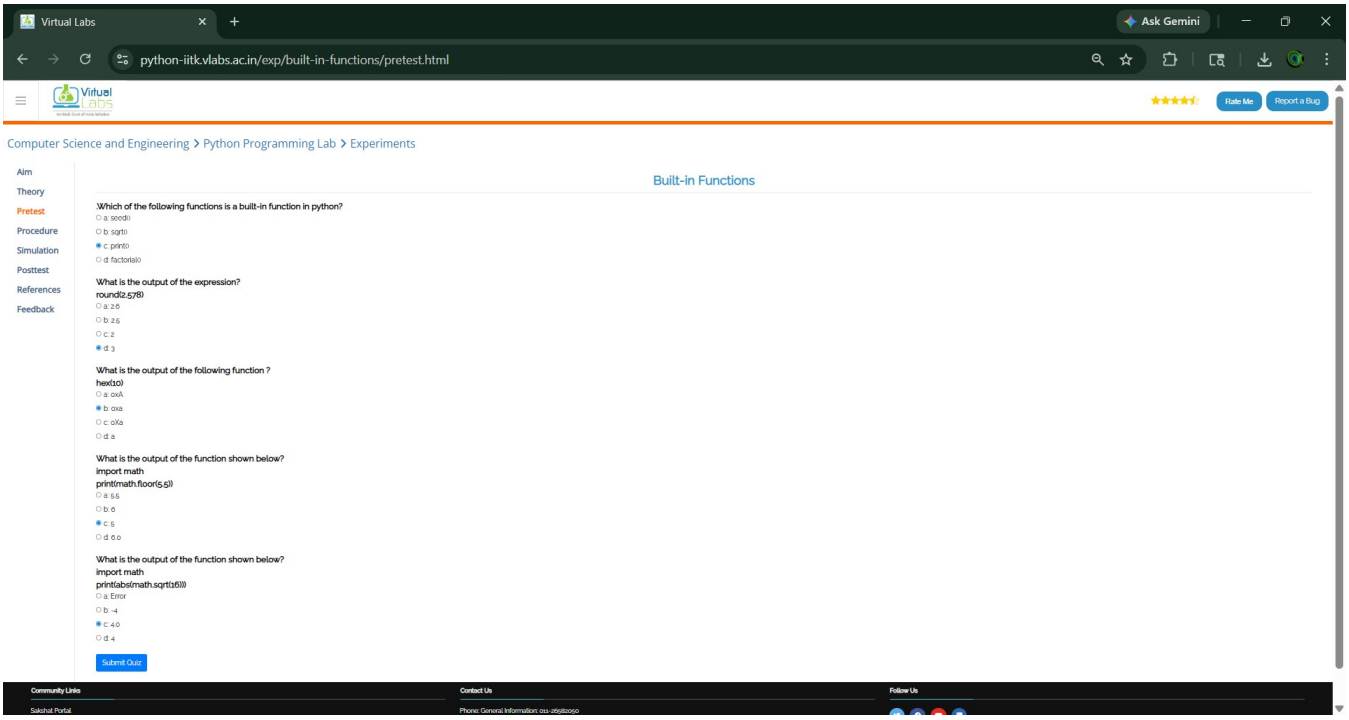
What will be the output of the following code?

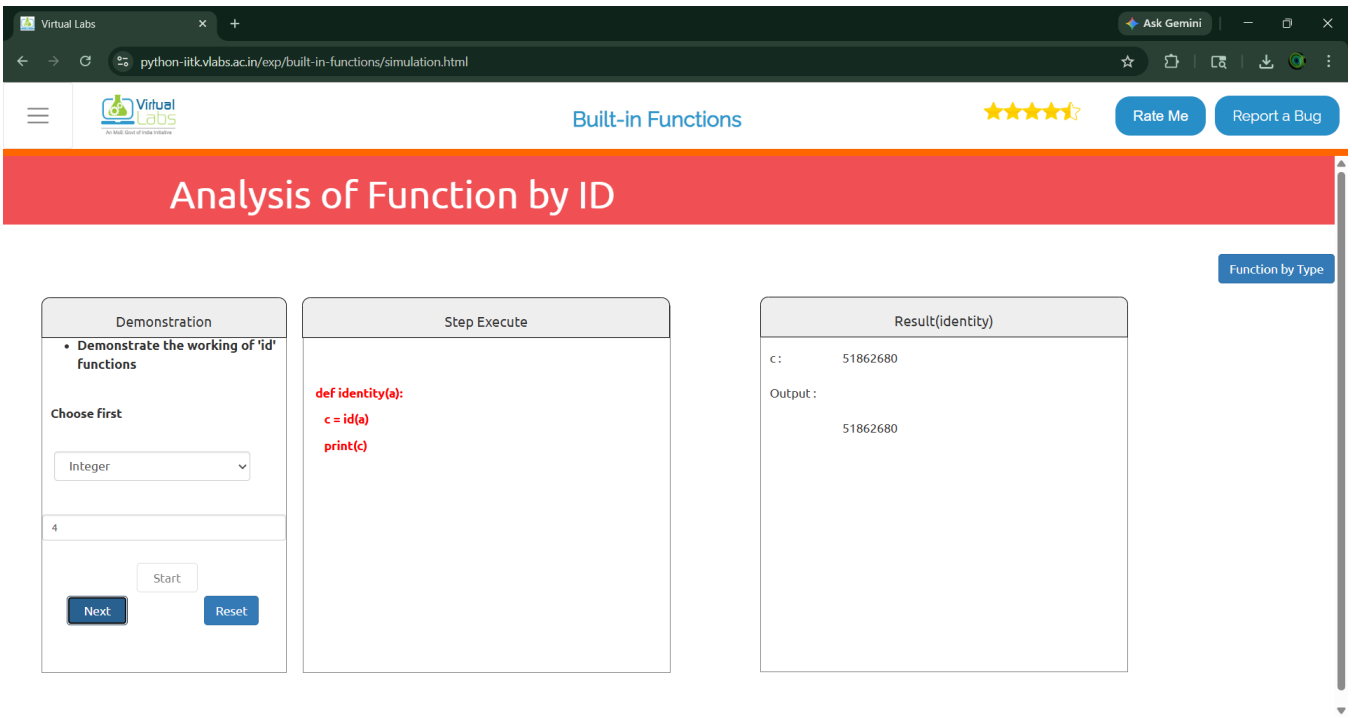
```
i=47
j=25
i=i%10
j=j%10
print(i*j)
```

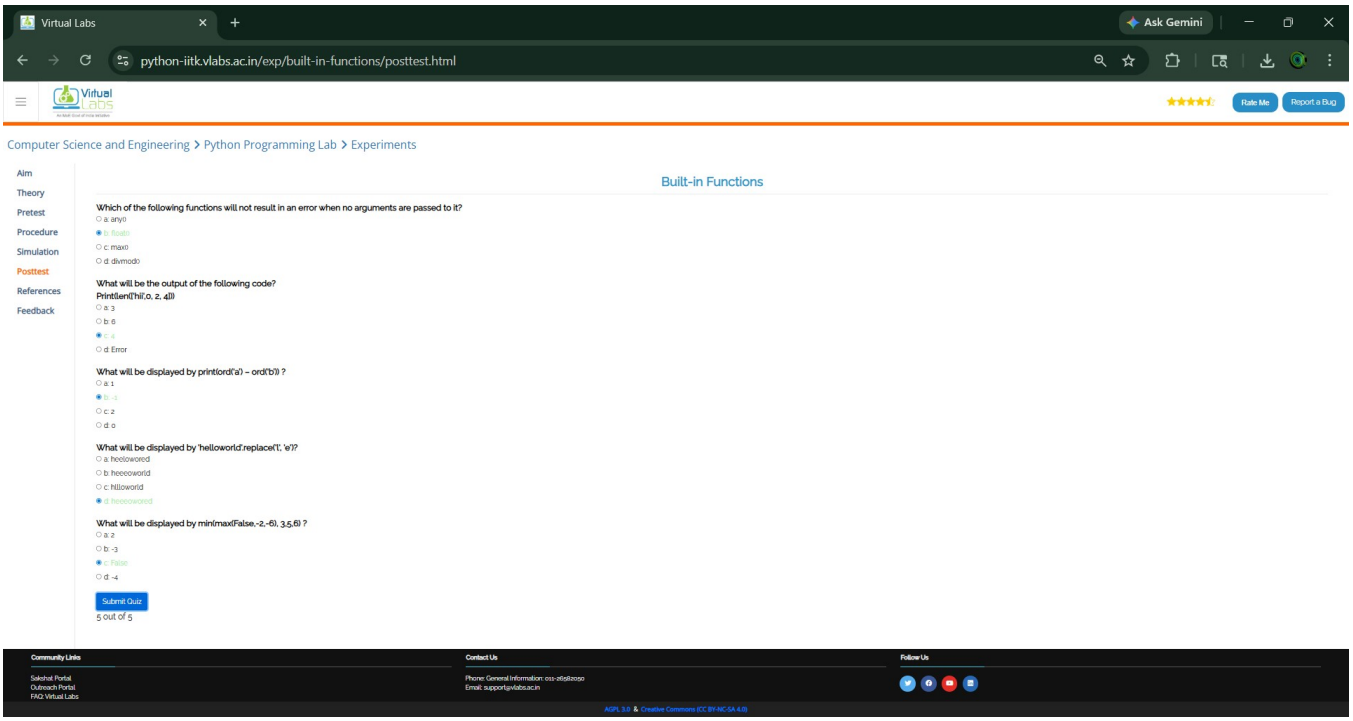
- a. 8
- b. 63
- c. 35
- d. 72

Thu, 26 Mar | 88% | ↓ 0.01 MB/s
10:54:41 PM | RAM 54% | CPU 16%

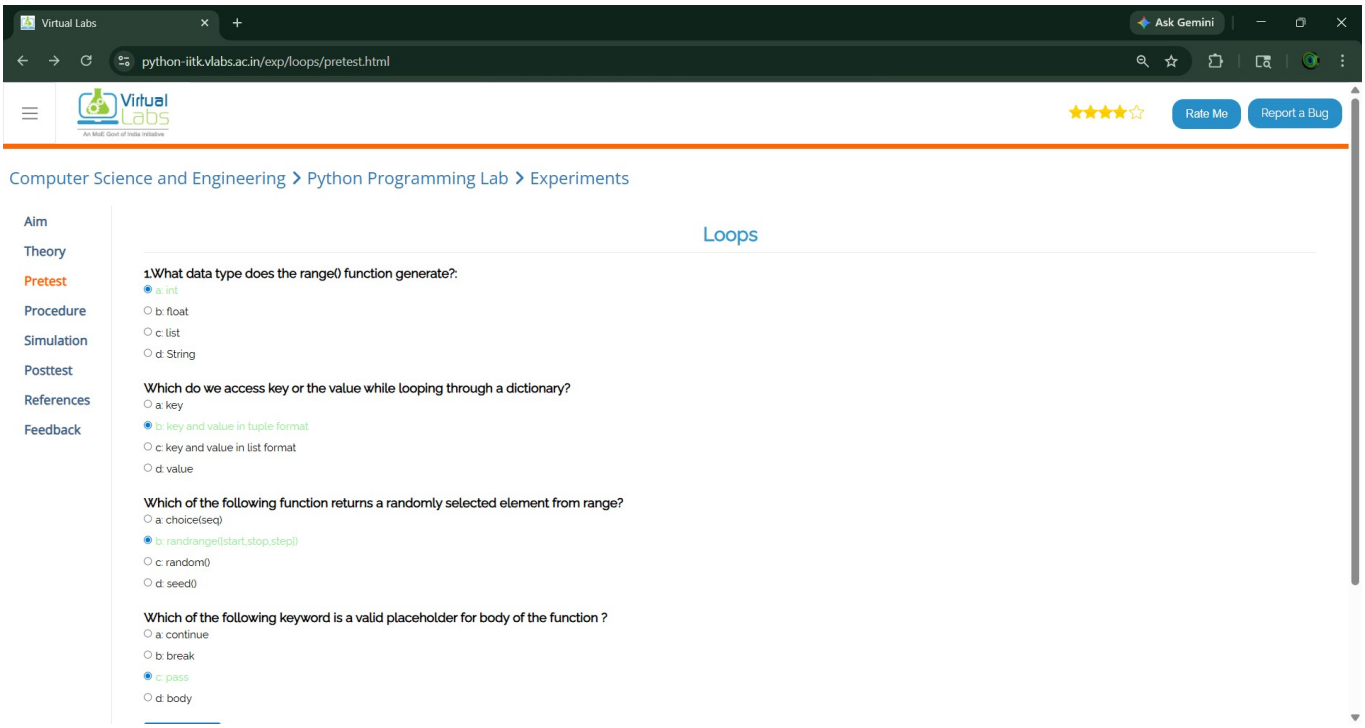
Built-in Functions

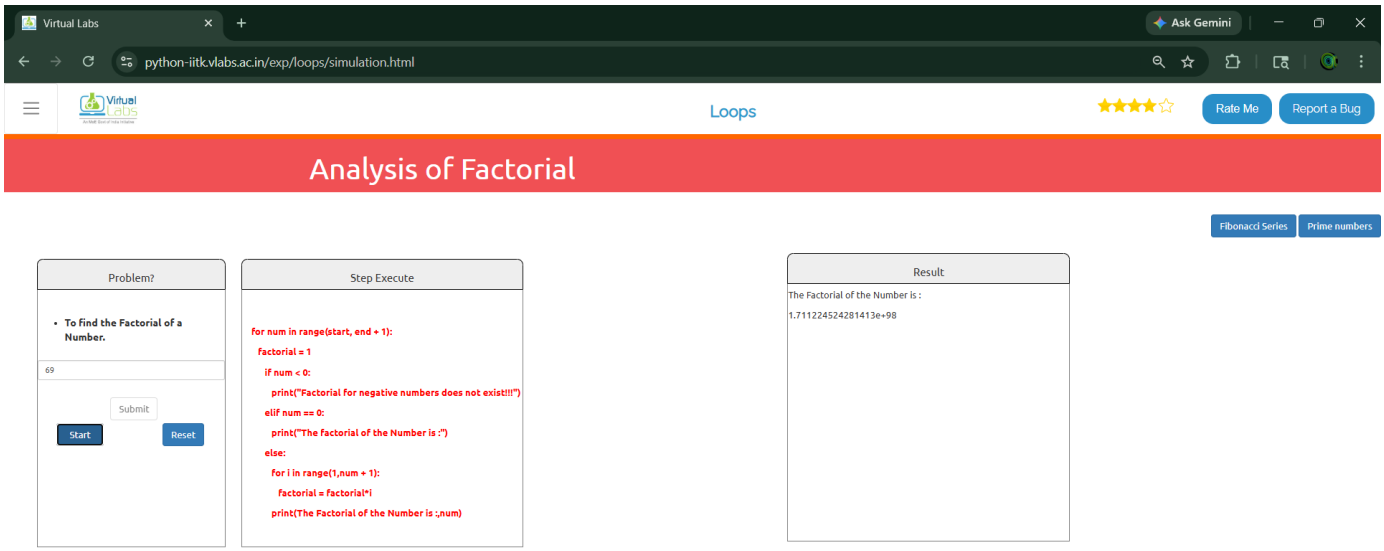


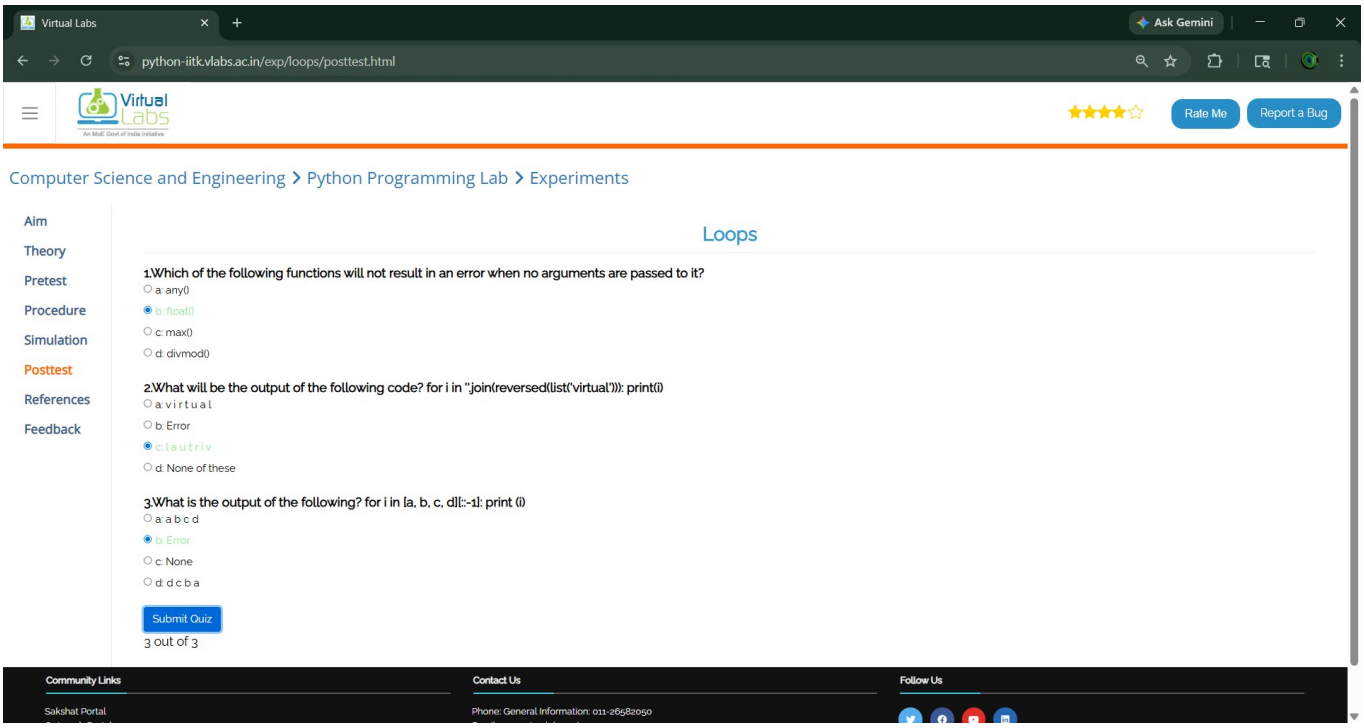



Virtual Labs
Ask Gemini
python-iitk.vlabs.ac.in/exp/built-in-functions/posttest.html
Rate Me
Report a Bug
Computer Science and Engineering > Python Programming Lab > Experiments
Aim
Theory
Pretest
Procedure
Simulation
Posttest
References
Feedback
Built-in Functions
Which of the following functions will not result in an error when no arguments are passed to it?
a: any()
b: float()
c: max()
d: divmod()
What will be the output of the following code?
Print(len('hi',0, 2, 4))
a: 3
b: 6
c: 4
d: Error
What will be displayed by print(ord('a') – ord('b')) ?
a: 1
b: -1
c: 2
d: 0
What will be displayed by 'helloworld'.replace('l', 'e')?
a: heelowored
b: heeeoworld
c: hlllloworld
d: heeeoworld
What will be displayed by min(max(False,-2,-6), 3,5,6) ?
a: 2
b: -3
c: False
d: -4
Submit Quiz
5 out of 5
Community Links
Sakshat Portal
Outreach Portal
FAQ: Virtual Labs
Contact Us
Phone: General Information: 011-26582050
Email: support@vlabs.ac.in
Follow Us
AGPL 3.0 & Creative Commons (CC BY-NC-SA 4.0)

Loops


Virtual Labs
python-iitk.vlabs.ac.in/exp/loops/pretest.html
Ask Gemini
Rate Me
Report a Bug
Computer Science and Engineering > Python Programming Lab > Experiments
Aim
Theory
Pretest
Procedure
Simulation
Posttest
References
Feedback
Loops
1.What data type does the range() function generate?:
a: int
b: float
c: list
d: String
Which do we access key or the value while looping through a dictionary?
a: key
b: key and value in tuple format
c: key and value in list format
d: value
Which of the following function returns a randomly selected element from range?
a: choice(seq)
b: randrange([start,stop,step])
c: random()
d: seed()
Which of the following keyword is a valid placeholder for body of the function ?
a: continue
b: break
c: pass
d: body


Virtual Labs
python-iitk.vlabs.ac.in/exp/loops/simulation.html
Ask Gemini
Loops
Rate Me
Report a Bug
Analysis of Factorial
Fibonacci Series
Prime numbers
Problem?
To find the Factorial of a Number.
69
Submit
Start
Reset
Step Execute
for num in range(start, end + 1):
factorial = 1
if num < 0:
print("Factorial for negative numbers does not exist!!!")
elif num == 0:
print("The factorial of the Number is :")
else:
for i in range(1,num + 1):
factorial = factorial*i
print(The Factorial of the Number is :,num)
Result
The Factorial of the Number is :
1.711224524281413e+98


Virtual Labs
python-iitk.vlabs.ac.in/exp/loops/posttest.html
Ask Gemini
Virtual Labs
Rate Me
Report a Bug
Computer Science and Engineering > Python Programming Lab > Experiments
Aim
Theory
Pretest
Procedure
Simulation
Posttest
References
Feedback
Loops
1.Which of the following functions will not result in an error when no arguments are passed to it?
a: any()
b: float()
c: max()
d: divmod()
2.What will be the output of the following code? for i in ''.join(reversed(list('virtual'))): print(i)
a: v i r t u a l
b: Error
c: l a u t r i v
d: None of these
3.What is the output of the following? for i in [a, b, c, d][::-1]: print (i)
a: a b c d
b: Error
c: None
d: d c b a
Submit Quiz
3 out of 3
Community Links
Contact Us
Follow Us
Sakshat Portal
Phone: General Information: 011-26582050

STUDENT DECLARATION

I hereby declare that the Virtual Lab experiments included in this document have been performed by me as part of my academic activity. The screenshots submitted are genuine and taken during the actual execution of the experiments.

Student Signature:

Date: 26 March 2026